NOT MEASUREMENT SENSITIVE

KKK-S-002769A
December 15, 1992
USED IN LIEU OF
KKK-S-2769
April 30, 1992

FEDERAL SPECIFICATION

SEMITRAILERS, STAKE: MODIFIED COMMERCIAL

This specification is approved for use by the US Army Tank-Automotive Command based upon currently available technical information but it has not been approved for promulgation as a coordinated revision of KKK-S-2769. It is subject to modification. However, pending its promulgation as a coordinated federal specification, it may be used in acquisition.

1. SCOPE

1.1 *Scope*. The specification covers modified commercial stake type semitrailers. Vehicles procured under this specification are commercial items that are required to be warranted by the manufacturer as specified in acquisition documents.

1.2 Classification. The semitrailer shall be one of the following types, sizes, and classes, as specified (see 6.2):

Type I - 10 900 kilograms (kg) (12 ton), single axle
Size C - 9140 millimeters (mm) (30 foot)

Type II - 18 150 kg (20 ton), tandem axle
Size D - 9750 mm (32 foot)
Size E - 10 660 mm (35 foot)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Tank-Automotive Command, ATTN: AMSTA-UED, Warren, MI 48397-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A

FSC-2330

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Size F - 12 200 mm (40 foot)
Size G - 13 700 mm (45 foot) 1/
Size H - 14 630 mm (48 foot) 1/

Type III - 22 700 kg (25 ton), tandem axle
Size F - 12 200 mm (40 foot)
Size G - 13 700 mm (45 foot) 1/
Size H - 14 630 mm (48 foot) 1/

Type IV - 30 800 kg (34 ton), tri-axle 1/
Size F - 12 200 mm (40 foot)
Size G - 13 700 mm (45 foot) 1/
Size H - 14 630 mm (48 foot) 1/

Class 1 - Slat sides
Class 2 - Panel sides
Class 3 - Flatbed, with stake pockets, but without slats or panels.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS
FEDERAL
CCC-C-419 - Cloth, Duck, Cotton, Unbleached, Plied Yarn, Army and Numbered.

MILITARY
MIL-P-514 - Plates, Identification, Instruction and Marking, Blank.
MIL-H-3912 - Hardwood Components, Fabricated: For Military Vehicles.
MIL-C-20696 - Cloth, Coated Nylon, Waterproof.

STANDARDS
FEDERAL
FED-STD-297 - Rustproofing of Commercial (Nontactical) Vehicles.

1/ May not be fully highway legal in some States.

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OCCUPATIONAL SAFETY AND HEALTH ADMINISTRATION (OSHA)
Code of Federal Regulations, 29 CFR.

(Application for copies of OSHA publications should reference the Code of Federal Regulations, 29 CFR, and the Federal Register and should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)
ASME Boiler and Pressure Vessel Code (BPVC)
Section IX - Qualification Standard for Welding and Brazing Procedures, Welders, Brazers, and Welding and Brazing Operators.

(Application for copies of ASME publications should be addressed to the American Society of Mechanical Engineers, United Engineering Center, 345 East 47th Street, New York, NY 10017.)

AMERICAN WELDING SOCIETY (AWS)
AWS B2.1 - Standard for Welding Procedure and Performance Qualifications.
AWS D1.1 - Structural Welding Code-Steel.

(Application for copies of AWS publications should be addressed to the American Welding Society, 2501 NW 7th Street, Miami, FL 31025.)

THE EUROPEAN TYRE AND RIM TECHNICAL ORGANISATION (ETRTO)
Standards Manual.

(Application for copies of ETRTO publications should be addressed to the European Tyre and Rim Technical Organisation, 32, Avenue Brugmann, 1060 Brussels, Belgium.)

INTERNATIONAL ORGANIZATION FOR STANDARDIZATION (ISO)
ISO 668 - Series I Freight Containers - Classification, Dimensions and Ratings.
ISO 1161 - Series I Freight Containers - Corner Fittings - Specification.

(Application for copies of ISO publications should be addressed to Global Engineering Documents, 7730 Carondelet, Suite 407, St. Louis, MO 63105.)

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MILITARY

MIL-STD-209	- Slinging and Tiedown Provisions for Lifting and Tying Down Military Equipment.
MIL-STD-1223	- Nontactical Wheeled Vehicles Treatment, Painting, Identification Marking and Data Plate Standards.
MIL-STD-1595	- Qualification of Aircraft, Missile and Aerospace Fusion Welders.
MIL-STD-1791	- Designing For Internal Aerial Delivery In Fixed Wing Aircraft.
MS 75021	- Connector, Receptacle, Electrical - 12-Contact, Intervehicular, 28-Volt, Waterproof.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, Military Specifications and Standards, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

AFSC DESIGN HANDBOOK

DH-1-11	- Air Transportability Handbook.

(Application for copies of the AFSC Design Handbook should be addressed to Wright-Patterson AFB, ASD/ENESS, Wright-Patterson AFB, OH 45433-6503, telephone (513) 255-6295.)

DEPARTMENT OF DEFENSE

Department of Defense Index of Specifications and Standards (DODISS).

(Copies of the DODISS are available on a yearly subscription basis either from the Government Printing Office for hard copy, or microfiche copies are available from the Director, Navy Publication and Printing Service Office, 700 Robbins Avenue, Philadelphia, PA 19111-5093.)

DEPARTMENT OF TRANSPORTATION (DoT)

Federal Motor Carrier Safety Regulations (FMCSR).
Federal Motor Vehicle Safety Standards (FMVSS).

(Application for copies of DoT publications should reference the Code of Federal Regulations, 49 CFR, and the Federal Register, and should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.)

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SAE, INC.
SAE Standards and Recommended Practices

J318	- Air Brake Gladhand Service (Control) and Emergency (Supply) Line Couplers - Trucks, Truck-Tractors, and Trailers (DoD adopted).
J560	- Seven-Conductor Electrical Connector for Truck-Trailer Jumper Cable.
J682	- Rear Wheel Splash and Stone Throw Protection (DoD adopted).
J700	- Upper Coupler Kingpin - Commercial Trailers and Semitrailers.
J702	- Brake and Electrical Connection Locations - Truck Tractor and Truck Trailer.
J1292	- Automobile, Truck, Truck-Tractor, Trailer and Motor Coach Wiring.

(Application for copies of SAE publications should be addressed to SAE, Inc., 400 Commonwealth Drive, Warrendale, PA 15096.)

THE TIRE AND RIM ASSOCIATION, INC.
Year Book.

(Application for copies of Tire and Rim Association publications should be addressed to The Tire and Rim Association, Inc., 175 Montrose West Ave., Copley, OH 44321.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Standard vehicle and accessories. Except as specified in 3.1.1 through 3.1.1.11.4, the vehicle, components, assemblies, and accessories to be delivered under the contract, shall be the manufacturer's standard or optional items, sold commercially to the public, which meet or exceed the requirements of this specification. All vehicle items shall be as represented and rated in the vehicle and equipment manufacturer's technical data. Technical data shall be limited to specifications and technical material, identical to that furnished to the authorized company representatives for selection of vehicle models and components and shall be available to the engineering offices of the procuring activity, prior to

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delivery of the items. The vehicle shall be a standard commercial semitrailer, of a model sold to a significant number of buyers other than the Government, in the course of normal business operations.

3.1.1 Special requirements. In addition to the standard vehicle and components specified in 3.1, the vehicle shall be furnished with special equipment as specified herein.

3.1.1.1 Treatment and painting. The vehicle shall be treated and painted in accordance with MIL-STD-1223. As specified by the procuring activity for the appropriate military service (see 6.2), the exterior color shall be in accordance with MIL-STD-1223.

3.1.1.2 Marking and data plates. As specified by the procuring activity for the appropriate military service (see 6.2), identification marking and data plates shall be in accordance with MIL-STD-1223.

3.1.1.3 Rustproofing. The vehicle shall be rustproofed in accordance with FED-STD-297.

3.1.1.4 Brake lights. At least one pair of brake lights shall override the four-way emergency flasher or the two systems shall be independent of each other. Modifications to the manufacturer's standard product to accommodate this requirement shall not compromise conformance to any Federal Motor Carrier Safety Regulation referenced herein or to any Federal Motor Vehicle Safety Standard. If additional lights are added to the vehicle, the lights shall be selected from the semitrailer manufacturer's standard matching hardware.

3.1.1.5 Welders and welding. In addition to the requirements of 3.23.4, all welders employed in the fabrication of the semitrailer shall be certified, before any welding is accomplished, in accordance with MIL-STD-1595; the welding qualifications of the ASME BPVC, Section IX; or AWS B2.1 or D1.1. The certification that the welders have passed the qualifications test shall be on file at the contractor's facility and shall be available for review by the Government.

3.1.1.6 Lubrication chart. An illustrated or schematic diagram type lubrication chart shall be provided on the semitrailer. The chart shall direct attention to all lubrication fittings and specify the range and grade of lubricants required for critical temperatures. The chart shall be permanently attached to the semitrailer in a readily visible location. The chart shall be inscribed on a plate conforming to composition A (class 1 or 2) or composition C of MIL-P-514.

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3.1.1.7 Hardwood. The platform flooring (see 3.16.3) shall be solid stock apitong, white or red oak, or white ash as specified in MIL-H-3912. Apitong is allowed for flooring only, provided the average weight shall be not less than 753 kilograms per cubic meter (kg/m^3) (47 pounds per cubic foot (pcf)), with none of the individual pieces weighing less than 673 kg/m^3 (42 pcf), with no more than 15 percent moisture content at time of weighing. Flooring is to be Government source inspected. Wood racks (stakes and slats) (see 3.17 through 3.17.6) shall be solid or jointed stock hardwood, other than apitong, conforming to MIL-H-3912. Plywood panel sides (see 3.18.1) need not conform to MIL-H-3912.

3.1.1.8 Wood treatment. Wood, including apitong, shall be treated with wood preservative in accordance with MIL-STD-1223.

3.1.1.9 24-volt lighting.

3.1.1.9.1 Independent 24-volt dc system. When specified (see 6.2), in addition to the standard 12-volt direct current (dc) lighting system (see 3.13 through 3.13.2), a 24-volt dc lighting system, independent of the 12-volt dc lighting system, shall be provided. Separate wiring and bulbs shall be provided for the 24-volt dc system. The system shall conform to 3.13. The 24-volt dc system shall also conform to the Federal Motor Carrier Safety Regulations referenced in 3.13.1, except the 24-volt dc stop and turn lamps shall be combined.

3.1.1.9.2 Interconnected 24-volt dc system. When specified (see 6.2), capability of operating the normal 12-volt dc system from a towing vehicle equipped with a nominal 24-volt dc electrical system shall be provided.

3.1.1.9.3 Receptacle for 24-volt dc system. When either an independent or interconnected 24-volt dc system is specified, the front of the semitrailer shall also be equipped with a 12-contact receptacle and cover conforming to MS 75021, part number MS 75021-1. The receptacle shall be located in accordance with SAE J702. The 24-volt dc, 12-contact receptacle shall be connected to the lights as follows:

Contact B. Connect to left-hand turn signal and stop lamp (yellow)
Contact D. Connect to ground (white)
Contact E. Connect to clearance, side marker, identification, and tail lamps (black and brown)
Contact J. Connect to right-hand turn signal and stop lamp (green).
Contact L. Connect to ground (white).

The remaining contacts shall not be connected. Circuits B and J on tactical (military design) trucks are combination stop and turn indicator circuits. On the interconnected 24-volt dc system, the normal 12-volt dc turn signal lights will function both as turn signals and stop lights, and the normal 12-volt dc stop lights will not be operational when the semitrailer is

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connected to a towing vehicle with a 24-volt dc power supply. Because of this condition, the stop light (red) circuit is not connected to the 24-volt, 12-contact receptacle.

3.1.1.9.4 Resistors. When an interconnected 24-volt dc system is specified, the 12-contact receptacle shall be provided with resistance in each circuit to reduce the voltage of a tactical towing vehicle from a nominal 28 volts dc to 12 volts dc. Each circuit resistor shall be selected to reduce the voltage of the tactical truck to within the maximum rated voltage of the semitrailer electrical components. The resistor assembly shall be located in a protective housing and provided with adequate ventilation or a heat sink to prevent overheating and any heat damage to resistors, wiring, or adjacent components.

3.1.1.10 Lifting and tiedown attachments. The semitrailer shall be equipped with lifting and tying down attachments. Lifting and tying down attachments shall conform to type II or type III of MIL-STD-209. A transportation plate conforming to composition A (class 1 or 2) or composition C of MIL-P-514 shall be attached to the semitrailer. The transportation plate shall be inscribed with a diagram showing the lifting attachments and proposed lifting slings, the capacity of lifting attachments and proposed lifting slings, the capacity of each attachment, and the required length and size of each sling cable. A silhouette of the semi-trailer, showing the center of gravity, shall be provided on the transporta-tion plate. Tiedown attachments shall be identified by stenciling or other suitable markings. Tiedown markings shall clearly indicate that the attachments are intended for the tiedown of the semitrailer on the carrier when shipped.

3.1.1.11 Air transportability. The vehicle shall be air transportable in accordance with MIL-STD-1791 and AFSC Design Handbook DH-1-11 in types C-130, C-141 and C-5 aircraft and shall be air transportable in type C-17 aircraft. Preliminary design guidelines for the type C-17 are 3610 mm (142 inches) maximum height, 5180 mm (204 inches) maximum width, 20 630 mm (812 inches) maximum length, 9 degree ramp angle, and floor load capacity equal to that allowed for the type C-5 aircraft. The semitrailer shall be air transportable without shoring, except for contractor designed and supplied shoring for under the landing gear wheels or pads. The semitrailer shall be air transportable both while coupled to a Government truck tractor selected by the contractor and while resting on the landing gear after being successfully backed aboard the aircraft by the contractor selected Government truck tractor.

3.1.1.11.1 Removable components. Removal or relocation of mechanically attached (nonwelded, nonriveted, etc.) components with common hand tools, requiring not more than 1 manhour total to remove, relocate and tiedown; and not more than 1 manhour total to return the vehicle to its original, as opposed to reduced, configuration; shall be acceptable. The stability and

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mobility of the vehicle shall not be affected by reducing its configuration. Tiedowns for removed or relocated equipment (eg. the racks) shall be furnished.

3.1.1.11.2 Air transportability verification. Three copies of comprehensive scale drawings satisfactorily showing each stage of loading, tiedown and unloading, for each of the aircraft specified above, will be required to be furnished to the engineering offices of the procuring activity, TACOM, AMSTA-UED, for subsequent review and approval of responsible Air Force Officials at Headquarters, Aeronautical Systems Division (AFSC), Wright-Patterson Air Force Base.

3.1.1.11.3 Scale drawings. Scale drawings shall include plan, side and end views of the semitrailer and a Government truck tractor selected by the contractor and outlines of the semitrailer and truck tractor negotiating required ramps and clearing required overhead door openings. As a minimum, the following information shall be shown:

MAIN FRAME
- (a) Forward and rear overhang
- (b) Height under front and rear overhang and at breakover angle
- (c) Landing gear pad size.

WHEEL ASSEMBLIES
- (a) Bogie articulation
- (b) Number of axles
- (c) Axle spacing
- (d) Number of wheels per axle
- (e) Tire size and inflation pressure
- (f) Tire load range (ply rating)
- (g) Tire footprint area.

WEIGHT/CAPACITY
- (a) Weight in shipping configuration
- (b) Weight distribution on each axle and at kingpin
- (c) Weight distribution on each axle and at landing gear
- (d) Number of tiedowns with accompanying capacity and location.

MANUFACTURER'S RATING DATA
- (a) Payload rating
- (b) Axle ratings
- (c) Suspension ratings
- (d) Landing gear rating
- (e) Tire ratings.

MISCELLANEOUS
- (a) Vehicle nomenclature and semitrailer's trade name, if any
- (b) Manufacturer's model number

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(c) Manufacturer's name
(d) Point of contact at the semitrailer manufacturer
(e) Telephone number at the semitrailer manufacturer
(f) Critical characteristics of the contractor selected Government truck tractor to be used to load the semitrailer on the aircraft.

3.1.1.11.4 Air transportability data plate. An air transportability data plate conforming to MIL-P-514, composition A (class 1 or 2) or composition C, shall be provided. The plate shall contain at least the following information:

(a) Side and rear silhouette views of the semitrailer
(b) Horizontal and vertical locations of the center of gravity of the semitrailer in transport condition
(c) Shipping weight
(d) Overall length, width and height
(e) Tiedown instructions for air transport in types C-130, C-141, C-5 and C-17 aircraft
(f) Required critical characteristics of the truck tractor to be used to load the semitrailer on the aircraft.

3.2 General design.

3.2.1 Federal Motor Vehicle Safety Standards. The vehicle shall comply with all Federal Motor Vehicle Safety Standards applicable to the type of semitrailer furnished and in effect on the date of manufacture.

3.2.2 OSHA and FMCSR regulations. The vehicle, with only furnished attachments, accessories and equipment, shall enable compliance by any user with all OSHA and FMCSR user regulations that are vehicle or vehicle equipment dependent. Regulations shall be those that are applicable to a commercial user of such a similar type, class and size of semitrailer for the same general use. See 6.1.

3.2.3 Ratings. Vehicle ratings shall be the manufacturer's published ratings. Component and vehicular ratings shall not be raised to meet the requirements of this specification. When published ratings are not available, verification of ratings shall be available to the appropriate engineering office of the procuring activity.

3.2.4 Dissimilar metals. All dissimilar metals used throughout the vehicle shall be insulated from one another to prevent galvanic or electrolytic action.

3.2.5 Prohibited materials. Asbestos materials shall not be used in any form in any part of the vehicle. No item, part or assembly shall contain radioactive materials in which the specific activity is greater than 0.002 microcurie per gram or activity per item equals or exceeds 0.01 microcuries.

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3.2.6 Accessibility. The design of the vehicle and optional equipment shall permit access for routine servicing and shall permit access for replacement and adjustment of component parts and accessories with minimal disturbance of other components and systems.

3.2.7 Lubrication. Lubrication means shall be provided for all parts of the semitrailer normally requiring lubrication. Where the use of high lubricating pressure will damage grease seals or other parts, fittings with pressure release shall be used.

3.2.8 Safety. All equipment or exposed portions of the equipment which are subject to extreme temperatures and inclement weather and all rotating or reciprocating parts which are of such a nature or so located as to become a hazard to operating personnel shall be insulated, fully enclosed, or properly guarded.

3.3 Performance. The semitrailer shall evidence no part failure, deformation, permanent set or interference between parts when towed, both empty and when loaded as specified in 3.4:

(a) At speeds as great as 32 kilometers per hour (km/h) (20 miles per hour (mph)) over unimproved roads and reasonably hard uneven terrain;
(b) At speeds as great as 97 km/h (60 mph) over improved roads; and
(c) At speeds as great as 16 km/h (10 mph) on 10 percent side slopes on unimproved roads and reasonably hard uneven terrain.

3.3.1 Turning ability. The semitrailer shall be capable of assuming a 90 degree angle to a coupled truck tractor without cramping and without damaging the semitrailer or the truck tractor.

3.3.2 Tracking ability. The semitrailer shall conform to the tracking requirements of Federal Motor Carrier Safety Regulation 393.70(a) when operated under the conditions specified in 3.3.

3.3.3 Brake performance. The service brakes shall stop the truck tractor-semitrailer combination, with the semitrailer loaded with specified payload, within the stopping distance requirements of Federal Motor Carrier Safety Regulation 393.52. Requirements shall be met with a truck tractor loaded to its full rated gross vehicle weight (GVW) and gross combination weight (GCW).

3.4 Payload and payload distribution. The rated payload capacity, evenly distributed over the length of the platform, shall be not less than that specified in table I.

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TABLE I. Rated payload and maximum wheel loading.

Type	Rated payload (not less than) kg	pounds	Total axle load (not more than) kg	pounds
I	10 400	(24,000)	9070	(20,000)
II	18 150	(40,000)	15 400	(34,000)
III	22 700	(50,000)	15 400	(34,000)
IV	30 800	(68,000)	See 3.7	

3.4.1 Total axle load. Except for type IV tri-axle semitrailer, the axle(s) shall be positioned so that the total load imposed on the tires, measured at the ground, with the semitrailer loaded as specified in 3.4, shall be not more than that specified in table I. For type IV tri-axle semitrailers, see 3.7.

3.5 Dimensions and clearances. The semitrailer, uncoupled from the towing vehicle, resting level on the landing legs, without payload, on level ground, shall conform to the dimensions specified in table II.

TABLE II. Dimensions and clearances.

Overall length (not less than)	As specified for applicable size in 1.2.
Overall width (not more than)	2440 mm (96 inches)
Inside width (not less than)	2240 mm (88 inches)
Ground clearance (not less than)	230 mm (9 inches)
Clearance under landing gear (not less than)	280 mm (11 inches) or greater if required for air transportability.
Landing gear swing clearance (radius from centerline of kingpin to any portion of semitrailer 150 mm (6 inches) or more below upper fifth wheel plate) (not less than)	2130 mm (84 inches)
Swing radius from centerline of kingpin to most distant point on semitrailer nose (not more than)	1330 mm (52.5 inches) for type I. 1500 mm (59 inches) for type II, III and IV.
Distance from centerline of kingpin to front end of semi-trailer (not less than)	610 mm (24 inches) for type I. 910 mm (36 inches) for type II, III and IV.
Distance from centerline of type I axle or type II or III rearmost axle to end of semitrailer (not more than), (For type IV, see 3.7)	1520 mm (60 inches)

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TABLE II. Dimensions and clearances. (continued)

Platform height (not more than) Rack, bulkhead and panel height (not less than) Upper fifth wheel height	1470 mm (58 inches) 1070 mm (42 inches) for type I. 1520 mm (60 inches) for type II, III and IV. 1220 ±25 mm (48 ±1 inch) for type I and II. 1330 ±25 mm (51 ±1 inch) for type III and IV.

3.6 Construction and stresses. Main frame members shall be fabricated of high strength steel with a yield strength of not less than 345 mega Pascals (mPa) (50,000 pounds per square inch). Main frame members shall be adequately braced with a sufficient number of crossmembers to present a rigid construction with sufficient strength for the dynamic and static loads imposed. The outside frame members shall run the full length of the load space. The rear crossmembers shall extend the full width of the semi-trailer. All parts of the framing shall be completely electric-arc welded. The strength of the finished welds shall be such that they will not fail or crack when the semitrailer is subjected to the loads and performance requirements specified herein. When the semitrailer is loaded as specified in 3.4, the maximum fiber stress in each frame member shall be not greater than 50 percent of the frame material yield strength or, when specified (see 6.2), 33 percent of the frame material yield strength. Under the above cited conditions of loading, the main frame deflection below horizontal shall be not greater than 50 mm (2 inches). The unloaded semitrailer may have a maximum of 50 mm (2 inches) of upward camber over the length of the platform.

3.6.1 Frame splices. Main frame members shall have one piece upper and lower flanges, runing the full length of the semitrailer. Where frame members are otherwise spliced, joints shall be reinforced to fulfill the dynamic and the static loading requirements specified herein. Splices shall be designed to avoid stress concentration. Splice welds shall be continuous. Frame depth transition areas, as in the gooseneck-to-over-fifth-wheel, shall be smooth and rounded to minimize stress concentrations.

3.6.2 Stress analysis. A stress analysis for the semitrailer, loaded as specified in 3.4, shall be performed. The stress analysis shall include shear and moment diagrams and deflection calculations. The strength of the floor, crossmembers, and outside frame members shall not be included in calculations to compute the main frame maximum fiber stress; but the weight of the floor, crossmembers, and outside frame members imposed on the main frame shall be included in the total load imposed on the main frame. Stress calculations shall include complete analysis of the frame depth transition to over the fifth wheel area, floor, crossmembers, longitudinal frame members and tiedown devices.

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3.7 Axles. Type I semitrailers shall have a single rear axle. Type II and III semitrailers shall have tandem rear axles. Type IV semitrailers shall have a rear tri-axle assembly. Axle ratings shall be at least equal to the load imposed on each axle, measured at the ground, with the semitrailer loaded as specified in 3.4. The tri-axle assembly on type IV semitrailer shall be positioned and individual axles spaced so not less than 3/5 of the semitrailer gross weight (curb weight plus payload) will be supported by the tri-axle assembly, with the semitrailer loaded as specified in 3.4. Further, the axles on type IV semitrailers shall be positioned and spaced to enable user compliance with the Surface Transportation Assistance Act of 1982, Federal Bridge Formula, Formula B or Table B, or when specified, the lesser allowable axle loads for the State specified (see 6.2). The wheel bearings and axle spindles on all types shall be oil lubricated. The oil viscosity shall be in accordance with the manufacturer's recommendations. The hubcaps shall have a window for visual determination of oil level. Provision for venting or other method of withstanding internal pressure buildup, and for replenishing the oil supply, shall be incorporated.

3.7.1 Sliding axle assembly. When specified (see 6.2), a sliding axle, spring and tire assembly shall be furnished. The sliding axle assembly shall have an adjustment length of not less than 2440 mm (96 inches) in increments of not more than 150 mm (6 inches). The arrangement shall permit the assembly to be manually unlocked, slid to a new position beneath the semitrailer by using the towing vehicle with the semitrailer brakes locked, and then manually relocking the assembly in the new position. The assembly shall be rated for the loads imposed with the semitrailer loaded as specified in 3.4.

3.8 Suspension system. Each component of the suspension system shall have a rated capacity at least equal to the load imposed, measured at the ground, with the semitrailer loaded as specified in 3.4. Clearances shall prevent interference between the tires and any other parts of the semitrailer under the operating conditions specified herein.

3.8.1 Full air suspension. When specified (see 6.2), the semitrailer shall be equipped with a full air suspension system, with components having a rated capacity at least equal to the load imposed on each member, measured at the ground, with the semitrailer loaded as specified in 3.4. Hydraulic double-acting shock absorbers shall be provided on the axle assembly. The suspension shall be provided with a mechanism at each wheel to assure lifting of the wheel and axle when jacking the semitrailer from the applicable jacking location. The suspension system shall incorporate leveling valves with time delay to avoid constant air consumption. Shock absorbers and springs shall be fabricated of materials capable of operating in ambient -46 degrees Celsius (^{o}C) (-50 degrees Fahrenheit (^{o}F)) to +52°C (+125°F). The suspension system shall have none of the sprung weight carried on anything other than the air springs. No other spring medium, such as leaf, coil, or rubber springs, shall be used in conjunction with the air springs.

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3.9 Wheels, rims and tires. The semitrailer shall have dual disc type wheels on each axle. The rims shall conform to Tire and Rim Association or European Tyre and Rim Technical Organisation recommendations for the type and size of tires furnished. Multi-piece rims shall not be furnished. Tire and rim sizes shall be the same for all wheels on the semitrailer.

3.9.1 Tires. Tires shall be steel belted radial, tubeless type, with highway tread. Tires shall be of rated capacity at least equal to the load imposed on each tire, measured at each wheel, at the ground, with the semitrailer loaded as specified in 3.4. Tires and tire ratings shall conform to Tire and Rim Association or to European Tyre and Rim Technical Organisation recommendations.

3.9.2 Spare tire assembly. The semitrailer shall have an inflated spare tire mounted on a spare wheel or rim, installed in a carrier. The spare tire shall be of the same size, tread design and load range (ply rating) as the tires in use on the semitrailer.

3.9.3 Spare tire carrier. A spare tire carrier shall be installed under the semitrailer, on the curbside, in an accessible location. Means shall be provided for securing the tire within the carrier to prevent pilferage and accidental loss. Threaded fasteners, when used to secure the spare tire in the carrier, shall be constructed of or plated with corrosion-resistant material. The carrier shall enable the safe removal and installation of the spare tire assembly from and to the carrier and ground without personnel positioning themselves or any part of their body under the spare tire assembly.

3.10 Brakes. The brakes shall be of the full air type and shall conform to Federal Motor Carrier Safety Regulations 393.40, 393.42, 393.43 and 393.45 through 393.47. Brake linings shall be of non-asbestos material. The braking system shall include:

(a) Automatic slack adjusters
(b) Piping
(c) Hose connections
(d) Glad hands
(e) Retracting rubber diaphragm or spring loaded dust covers or dummy glad hands equipped with security chains or cables
(f) Provisions for stowing dummy glad hands and
(g) All other components required for a complete air-brake system.

Glad hands shall conform to SAE J318. Glad hand location shall comply with SAE J702. The braking system shall be installed in a manner which provides road clearance for travel over uneven terrain and protection against damage caused by objects striking components. No part of the braking system shall extend below the bottom of wheel rims.

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3.10.1 Parking brakes. Spring or air diaphragm mechanical lock type parking brakes shall be provided. The parking brakes shall be automatically applied upon disconnection of the supply (emergency) air line and under emergency braking conditions. The parking brakes shall hold the semitrailer, with specified payload, on a 10 percent grade despite the depletion of the compressed air supply. The parking brake shall hold the loaded semitrailer headed either up or down the grade. Parking brakes shall conform to Federal Motor Carrier Safety Regulation 393.41.

3.11 Upper fifth wheel plate. The upper fifth wheel plate shall be designed for completely covering and for use with either a full oscillating or fore and aft rocking lower fifth wheel. The upper fifth wheel plate shall be of sufficient strength to be fully supported by a lower fifth wheel 910 mm (36 inches) in diameter and shall conform to Federal Motor Carrier Safety Regulation 393.70(b). The forward end of the upper fifth wheel plate shall have a turned up lip not less than 50 mm (2 inches) high for ease of coupling. The kingpin shall be of heat treated alloy steel and shall conform to SAE J700.

3.12 Landing gear. The semitrailer shall have two vertical lift, telescopic, nonrotating landing legs, with two speed gears and a handcrank on the curbside. Landing legs shall be equipped with not less than 230 mm (9 inch) diameter steel wheels or self-leveling sand shoes of not less than 250 mm by 250 mm (10 inch by 10 inch) square bearing area. Supports for the crank extension shaft and clips for holding the crank when folded shall be provided. The landing gear shall withstand, without deformation, the combined static and dynamic forces due to the proportion of the gross weight sustained and the forces resulting from impact during coupling and uncoupling operations. When placed in travel position, the landing gear legs shall remain positively locked. The landing gear shall be protected to prevent the entrance of foreign matter which would impair its functioning or mechanical efficiency. The maximum cranking effort to raise the fully loaded semitrailer shall be not more than 45 kg (100 pounds).

3.12.1 Landing gear range of adjustment. The landing gear shall have a range of adjustment sufficient to vary the height of the upper fifth wheel from 1190 mm (47 inches) to not less than 1320 mm (52 inches) from the ground. With the semitrailer coupled to a truck tractor and in level position, the clearance under the fully retracted landing gear shall be not less than as specified in table II.

3.12.2 Air/hydraulic powered landing gear. When specified (see 6.2), in addition to the requirements of 3.12 and 3.12.1 (less hand crank, etc.), the landing gear shall be air/hydraulic powered. The landing gear system shall include a control box equipped with a 12-volt dc motor driven pump, master control valve, leveling valve, booster, main oil tank, air side oil tank, key switch and mounting hardware.

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3.13 Lighting. All lights and reflectors shall be protected from operational hazards by mounting in recessed or otherwise guarded locations. Lights and reflectors shall not be mounted on vertical surfaces of the rub rail (unless recessed and fully protected) or on the semitrailer bumper. All electrical wiring shall conform to SAE J1292. Clearance, side marker and identification lamps shall be constructed for easy removal and replacement of lamps and lenses without the use of any tools.

3.13.1 Twelve-volt dc system. A 12-volt dc lighting system conforming to 3.13 and 3.13.2 shall be provided. The 12-volt dc lighting system shall conform to Federal Motor Carrier Safety Regulations 393.9, 393.11, 393.20, 393.22, 393.23, 393.25 through 393.29, 393.32 and 393.33.

3.13.2 Receptacle for 12-volt dc system. The front of the semitrailer shall be equipped with a 7-contact receptacle conforming to SAE J560 with the connectors connected and color coded as specified therein. The receptacle shall be provided with a spring loaded cover. The receptacle shall be located in accordance with SAE J702.

3.14 Rear end protection. Semitrailer rear end protection shall be in accordance with Federal Motor Carrier Safety Regulation 393.86. In addition, a dock bumper extending not less than 38 mm (1.5 inches) beyond the rear of semitrailer, shall be furnished. The dock bumper shall be the full width of the platform and shall be located approximately at floor level.

3.15 Mud flaps. Mud flaps shall be provided to the rear of the rear wheels, in accordance with SAE J682. A metal strip not less than 3 mm (0.125 inch) thick and not less than 25 mm (one inch) wide, extending the entire width of the mud flap, shall be installed to prevent the bolt heads or bolt nuts from damaging the mud flap. As an alternate method of attaching the mud flaps, tabs or clips with minimum surface contact dimensions of 25 mm (1 inch) high by 32 mm (1.25 inch) wide by 2.4 mm (0.094 inch) thick shall be furnished at each bolt.

3.16 Platform. The platform shall have crossmembers spaced on not more than 300 mm (12 inch) centers. The platform shall have a flat front with not more than 300 mm (12 inch) radius front corners or bevel front corners, both to provide not more than the maximum swing radius specified in table II. Bevel front corners shall extend not more than 300 mm (12 inches) to the rear from the front of the platform. The rear platform corners shall be square.

3.16.1 Stake pockets and rub rails. Protruding steel stake pockets shall be furnished. Stake pockets shall be protected on all four sides of the platform by steel rub rails not less than 50 mm (2 inches) wide and not less than 7.9 mm (0.3125 inch) thick. The rub rails shall be suitable for securing cargo or lashing D-ring tiedown devices for that purpose shall be provided.

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3.16.2 Lashing D-rings. When the rub rails are unsuitable for securing cargo, lashing D-rings for that purpose shall be furnished. The lashing D-rings shall not protrude beyond the outer edge of the platform rub rails when the D-rings are not in use. The semitrailer shall have lashing D-rings located as follows:

(a) Two D-rings on the front and two D-rings on the rear, equally spaced from the center of the platform
(b) On both sides of the platform, lashing D-rings equally spaced every 910 mm (three feet) for the length of the platform. Minor variations is spacing due to interference with other components of the semitrailer are acceptable.

Each lashing D-ring and its mounting shall be capable of withstanding a pull of 4540 kg (10,000 pounds) in all directions without permanent deformation.

3.16.3 Platform flooring. Flooring shall be of solid stock hardwood and shall conform to 3.1.1.7 and 3.1.1.8. Laminated hardwood shall not be furnished. Floorboards shall be not less than 32 mm (1.25 inches) thick and not less than 1820 mm (6-feet) long, laid lengthwise with shiplap joints. Butt joints shall be centered over crossmembers and shall be staggered.

3.16.4 Attachment of flooring. Flooring shall be secured with countersunk phosphate coated or other corrosion-resistant screws or bolts. Not less than two securing screws or bolts shall be installed at each cross sill and not less than two on each side of each butt joint. The top of countersunk screw or bolt heads shall be not less than 3.2 mm (0.125 inch) and not more than 6.4 mm (0.25 inch) below the top of the wood surface.

3.16.5 Container securement devices. When specified for size F and longer semitrailers (see 6.2), the platform shall have two sets of four ISO container securement devices. The securement devices shall be of the retractable type which shall provide for a flat platform when retracted and not in use. Eight ISO securement devices, four in each longitudinal half of the semitrailer platform, shall be provided. The securement devices shall be located for the alinement and securement of two ISO freight containers, "1 C" Designator, 6100 mm x 2440 mm x 2440 mm (20 feet x 8 feet x 8 feet), as specified in ISO 668, with freight container corner fittings conforming to ISO 1161. The securement devices shall be mounted with reinforcements as needed to meet the requirements of Federal Motor Carrier Safety Regulation 393.100(e).

3.17 Class 1 slat side semitrailer. Class 1 slat side semitrailer shall be furnished with side and rear racks and a removable front bulkhead, all of the height specified in table II.

3.17.1 Front bulkhead. The front bulkhead shall be of steel or aluminum. The front bulkhead shall completely cover the front of the semi-trailer, to the full width and height of the side racks (panels). The front

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bulkhead shall have reinforced or flanged edges. The front bulkhead shall be capable of withstanding a uniformly distributed horizontal static load of 1300 kg (3,000 pounds) without permanent distortion of the bulkhead or its mountings. The bulkhead shall be easily removable.

3.17.2 Side and rear racks. Individual rack section width shall not exceed 1370 mm (54 inches). All sections shall be easily removable and shall show no evidence of binding or jamming. The intermediate side sections shall be interchangeable on the same side, except for the adjusting sections. All sections shall meet and fit so that locking devices engage and disengage freely and shall not require the use of other than small hand tools.

3.17.3 Stakes. When the racks are installed, the slats shall be inside the loadspace, with the upright stake posts outside. The stakes shall be either hardwood conforming to 3.1.1.7 and 3.1.1.8, fitted with a steel ferrule at the lower end; pressed or structural steel sections of not less than 3.41 mm (0.1345-inch) (U.S. revised standard gage No. 10) steel; or extruded aluminum shapes of not less than 2.4 mm (0.094-inch) aluminum.

3.17.4 Slats. Slats shall be of hardwood conforming to 3.1.1.7 and 3.1.1.8. All slats shall be at least 19 mm by 89 mm (0.75 by 3.5 inches) in finished dimensions. The slats in the bottom half shall be spaced as close together as practicable, and the slats in the upper half shall be spread not less than 50 mm (2 inches) and not more than 75 mm (3 inches) apart. Slat edges and end corners shall be rounded.

3.17.5 Hardware. Stakes and slats, including hardware, shall be held together with bolts, rivets or other through-type similar metal (see 3.2.4) fasteners at least 6.4 mm (0.25 inch) in nominal diameter. Bolt or rivet heads shall be against the slats, on the inside of the loadspace. The front bulkhead and side and rear racks shall be provided with heavy-duty interlocking hardware. Each rack section shall be equipped with locking devices to lock each rack directly to the platform. Body hardware shall be attached to the rack slats with not less than four bolts or rivets for each piece of hardware.

3.17.6 Crosstie chains. Size C through size E semitrailers shall have two crosstie chains. Size F and longer semitrailers shall have three crosstie chains. Crosstie chains shall be installed at equidistant points across the semitrailer, fastened to the top of the racks (panels) on one end. The other end of each chain shall be fitted with a grab hook, to mate with a fastener to be provided at the top of the rack (panel) opposite the fixed end of the chain.

3.18 Class 2 panel side semitrailer. Class 2 panel side semitrailer shall be furnished with side and rear panels constructed of plywood, sheet steel, or aluminum alloy. Steel and aluminum panels shall be reinforced at the top, bottom and each end. Steel and aluminum panels shall be corrugated, ribbed or reinforced to provide adequate stiffness. Any wood reinforcements

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shall be plywood, treated as specified in 3.1.1.8. Individual panel sections shall not exceed 1370 mm (54 inches) in width. The intermediate side sections shall be interchangeable on the same side, except for the adjusting sections. The cross section configuration on the vertical panel edges shall be such that adjoining panels will interlock or overlap to minimize the entrance of water. The panel tops shall be parallel with the platform surface and positively fastened together for rigidity and alinement. Panels across the rear shall be provided. All panels shall be readily removeable and shall show no evidence of binding or jamming. Otherwise, the panels, bulkhead, stakes, hardware and chains shall conform to 3.17 through 3.17.6.

3.18.1 Plywood panels. When plywood panel construction is furnished, each side and rear panel assembly shall consist of stakes as specified in 3.17.3 and a plywood panel. The panels shall be exterior grade plywood of not less than 13 mm (0.50 inch) nominal thickness. Plywood panels shall be treated as specified in 3.1.1.8.

3.18.2 Steel panels. When steel panel construction is furnished, each side and rear panel assembly shall consist of steel stakes as specified in 3.17.3 and a sheet steel panel of not less than 2.65 mm (0.1046 inch (U.S. revised standard gage No. 12)) thickness. One end of adjoining panels shall be offset the thickness of the metal, to form a 25 mm (1 inch) overlap with the edge of the next panel, or other suitable means of supporting the joints and interlocking with the adjacent panels shall be used.

3.18.3 Aluminum panels. When aluminum panel construction is furnished, panel assemblies shall consist of 6061-T6 aluminum alloy not less than 32 mm (0.125-inch) thick; and extruded aluminum alloy hat shaped stake posts. One end of adjoining panels shall be offset the thickness of the metal, to form a 25 mm (1 inch) overlap with the edge of the next panel, or other suitable means of supporting the joints and interlocking the adjacent panels shall be used.

3.19 Class 3 flatbed semitrailer. Class 3 flatbed semitrailer shall be furnished with a front bulkhead, conforming to 3.17.1. Stake pockets, but not racks or panels, are required.

3.20 Roof bows, tarpaulin and tiedown ropes. When specified (see 6.2), class 1 slat side semitrailer shall be provided with roof bows, a fitted tarpaulin and tiedown ropes. Unless otherwise specified (see 6.2), class 2 panel side semitrailer shall be so equipped. The roof bows shall be spaced on centers of not more than 610 mm (24 inches), with bow curvature height not less than 360 mm (14 inches) above the top of the racks (panels).

3.20.1 Tarpaulin. The tarpaulin color shall harmonize with the vehicle exterior color. The tarpaulin shall be hard texture cotton duck cloth not less by weight per square yard than No. 10 cotton duck conforming to CCC-C-419; or vinyl coated nylon conforming to type II, class 2 of

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MIL-C-20696. The tarpaulin shall be treated or shall be inherently resistant to fire, water, weather and mildew. All seams shall be double stitched. Reinforcing pads of the size, shape, and in the positions required to protect the tarpaulin at the rack or panel corners and at the roof bows shall be sewn in the tarpaulin. The tarpaulin shall cover the entire body when spread over the roof bows and shall extend down the sides, front, and rear to not less than 75 mm (3 inches) below the middle of the racks or 75 mm (3 inches) below the top of the front bulkhead or side and rear panels, as applicable.

3.20.2 Tiedown ropes. Manila tiedown ropes, 9.5 mm (3/8-inch) in nominal diameter, shall be spliced through tarpaulin grommets spaced not more than 610 mm (2 feet) apart. The free end of each rope shall be secured with a metal clip or seized with waxed twine to prevent raveling. Tiedown ropes shall be of sufficient length to permit tying them to the platform rub rail or tiedown devices.

3.21 Tool box. The vehicle shall have a tool box. Minimum dimensions shall be 560 mm (22 inches) by 560 mm (22 inches) by 300 mm (12 inches). The door opening shall be not less than 510 mm (20 inches) by 250 mm (10 inches) in size. The tool box shall be made of not less than 2.657 mm (12 gage) (0.1046 inch) steel or of equivalent strength aluminum. The box shall be weatherproof and shall provide for locking with a padlock.

3.21.1 Tool box location. The tool box shall be underslung, on the curbside of the vehicle. The tool box shall not reduce the ground clearance or ramp breakover angle of the vehicle, as found before the tool box is mounted.

3.22 Servicing and adjusting. Prior to acceptance of the semitrailer by the Government inspector, the contractor shall service and adjust the semitrailer for immediate operational use. The servicing and adjustment shall include at least the following:

(a) Inflation of all tires
(b) Adjustment of brakes
(c) A check for proper functioning of all lights
(d) Complete lubrication with grades of lubricants recommended for ambient air temperature at the delivery point.

3.23 Workmanship. Defective components or parts and assemblies which have been repaired or modified to overcome deficiencies shall not be furnished. Bolted, riveted and welded construction used shall be in accordance with the highest standards of the industry.

3.23.1 Metal fabrication. Metal used in the fabrication of equipment shall be free from kinks and sharp bends. The straightening of material shall be done by methods that will not cause damage to the metal. Shearing and chipping shall be done neatly and accurately. All bends of a major character shall be made with controlled means in order to ensure uniformity of size and shape.

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3.23.2 Bolted connections. Bolt holes shall be accurately punched or drilled and shall have the burrs removed. Washers or lockwashers shall be provided in accordance with good commercial practice, and all bolts, nuts, and screws shall be tight.

3.23.3 Riveted connections. Rivet holes shall be accurately punched or drilled and shall have the burrs removed. Rivets shall be driven with pressure tools and shall completely fill the holes. Rivet heads, when not countersunk or flattened, shall be of approved shape and of uniform size for the same diameter of rivet. Rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member.

3.23.4 Welding. Welding procedures shall be in accordance with a nationally recognized welding code. The surface of parts to be welded shall be free from rust, scale, paint, grease or other foreign matter. Welds shall be of sufficient size and shape to develop the full strength of the parts connected by the welds. Welds shall transmit stress without permanent deformation or failure when the parts connected by the welds are subjected to proof and service loading. See 3.1.1.5.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility for ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Government verification. Quality assurance operations performed by the contractor will be subject to Government verification at unscheduled intervals. Verification will consist of observation of the operations to

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determine that practices, methods, and procedures of the contractor's inspection are being properly applied. Failure of the contractor to promptly correct product deficiencies discovered shall be cause for suspension of acceptance until correction has been made or until conformance of the product to specification criteria has been demonstrated.

4.3 First production vehicle inspection. The first vehicle of each type, size and class produced under the contract shall be inspected by the contractor at his plant under the direction and in the presence of Government representatives. The purpose of the inspection shall be to determine vehicle conformance with the requirements of the contract. Acceptance of the first production vehicle shall not constitute a waiver by the Government of its rights under the provisions of the contract.

4.3.1 Semitrailer weight. The semitrailer shall be weighed to determine its curb weight and the distribution of the curb weight on the kingpin and the rear axle(s). The imposed loading on the kingpin and rear axle(s), with payload, shall be computed using the curb weight and the semitrailer payload distributed as specified in 3.4. The rated capacity of the axles, suspension system and tires shall be compared to the loads imposed on them to determine if they conform to contract requirements.

4.3.2 Normal road test. Unless an extended road test is specified (see 4.3.3), the first vehicle for each contract line item number (CLIN) under contract shall be road tested with and without payload. The road test shall be for the distance required in table III, corresponding to the number of vehicles being procured under the specific contract line item number (CLIN). The test shall be conducted by the contractor under the direction of Government representatives. The test shall be conducted over a test course approved by the procuring activity and meeting the requirements of table IV. The vehicle shall be inspected periodically during and at the completion of the test by the contractor.

TABLE III. Road test distance.

Number of vehicles being procured	Road test distance km (miles)
1 - 10	160 (100)
11 - 20	320 (200)
21 - 30	480 (300)
31 - 40	640 (400)
41 - 50	800 (500)
51 - 100	1600 (1000)
101 - up	8050 (5000)

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TABLE IV. Test course requirements.

Type of road	Road test distance
Dirt	10 percent
Gravel	30 percent
Improved pavement	60 percent

4.3.2.1 Test course. The total test mileage shall be on roads and highways within the general geographic limits of the place of manufacture and adjoining states. Any special highway permits required shall be obtained by the contractor. The course shall be arranged to return the vehicle to the manufacturer's plant at the completion of each 1600 km (1,000 miles), if applicable, or at the completion of the test otherwise, for inspection and chassis servicing.

4.3.2.2 Test loads. The semitrailer shall be tested for 90 percent of the mileage with the semitrailer loaded as specified in 3.4 and for 10 percent of the mileage with the semitrailer empty. Average speeds during all road tests shall be as close as possible to those specified in 3.3.

4.3.2.3 Road test failure. The vehicle shall successfully complete the entire test. Failure of the vehicle to successfully complete the test shall be cause for nonacceptance of any of the contract quantity, pending correction of deficiencies and evidence of corrective action which shall prevent the recurrence of similar deficiencies in the contract quantity. Failure of the vehicle to successfully complete the test shall not constitute an excusable delay in meeting scheduled deliveries. The equipment shall be closely monitored throughout the test for interference with towing operations, and for instability as evidenced by poor tracking or uncontrollable sideslipping, skidding, swerving, or tilting. The equipment also shall be closely monitored for loosening of parts, interference between parts, leakage of fluids or lubricants, overheating of components, and for damage, permanent distortion, or excessive wear of parts and components. The equipment shall be disassembled to the extent necessary for inspection for unusual wear or damage to components. Substitution of a new semitrailer or replacement of a major component may require complete retest at the discretion of the Government representatives. Rejection of the test semitrailer shall be for damage or deficiencies, including but not limited to the following:

(a) Damage caused by collision
(b) Failure of any major component
(c) Vibration due to misalinement of wheels or frame
(d) Vibration due to the type of construction or mounting
(e) Evidence of abnormal tire wear due to misalinement or unbalance

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(f) Failure of any vehicular safety device such as brakes or electrical circuits
(g) Evidence of structural weakness in any part of the vehicles, vehicle components, accessories or welds
(h) Loose mountings of parts or accessories due to workmanship or vehicular operation.

4.3.3 Extended road test. When specified (see 6.2), the first vehicle shall be subjected to an extended road test as specified in table V. While negotiating the specified side slopes, in both directions, the equipment shall be subjected to not less than 10 sine wave turns. While negotiating the paved road course at highway speeds, the equipment shall be periodically subjected to at least 25 sudden stops at maximum braking effort and shall be periodically subjected to at least 25 left and 25 right turn rapid lane change maneuvers that simulate avoidance of collisions.

TABLE V. Extended road test.

Surface	Load condition	Average speed km/h (mph)	Distance km (miles)
Side slope, 10 percent	Full load	8 - 16 (5 - 10)	
	Half load	8 - 16 (5 - 10)	
	Empty	8 - 16 (5 - 10)	
Belgian block	Full load	16 - 32 (10 - 20)	480 (300)
	Half load	16 - 32 (10 - 20)	320 (200)
	Empty	16 - 32 (10 - 20)	160 (100)
Paved road	Full load	88 (55)	3380 (2100)
	Half load	88 (55)	2250 (1400)
	Empty	88 (55)	1130 (700)
Improved gravel road	Full load	40 - 56 (25 - 35)	1450 (900)
	Half load	40 - 56 (25 - 35)	970 (600)
	Empty	40 - 56 (25 - 35)	480 (300)

4.3.3.1 Patterning. High stress areas of the frame assembly shall be coated with brittle lacquer prior to the road test. The semitrailer shall be fully loaded and towed over the Belgian block test course for one cycle.

4.3.3.2 Strain recording. Strain gage rosettes shall be positioned in locations of high stress concentrations. The locations shall be as determined by the brittle lacquer patterning. A minimum of ten strain gages shall be used. Locations of strain gages shall include the gooseneck, the kingpin area and transverse areas. Strains shall be recorded for each of the following three cycles of the test course:

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(a)	Full load	1 cycle
(b)	40 to 60 percent of full load	1 cycle
(c)	No load	1 cycle

Strain/occurence histograms shall be supplied for each strain gage for each test cycle. The actual test course shall be recorded on the histogram. Pictures indicating the position of each strain gage shall be provided.

4.3.3.3 Evaluation. The ultimate, tensile and yield strengths of the gaged components shall be documented. Any strain recorded indicating stress greater than that specified in 3.6 shall be cause for rejection.

4.3.3.4 General design and performance characteristics verification. The equipment shall be verified as suitable for the intended use by satisfactorily completing the preceding inspections and tests. Equipment submitted for test shall contain all specified components and shall be properly adjusted and serviced for operation. Apparatus used in conjunction with the tests shall be calibrated for accuracy. In addition to the results of inspections and tests, test records shall include the following information, as applicable.

(a) Name, date(s), and location of test
(b) Description of test course and weather conditions that could affect test results
(c) Condition of the equipment at the start of test
(d) Tire inflation pressure
(e) Types of lubricants used
(f) Description of test instrumentation and recording apparatus, including calibration records
(g) Description of the truck tractor used
(h) Load condition of the equipment
(i) Average and maximum speeds
(j) Distance traveled, including total test distance and accumulated distance, and the time of each major incident such as component failure or adjustment
(k) Component failure analysis and corrective measures
(l) Component adjustment, and method and ease of adjustment
(m) Any noted safety hazards
(n) Condition of the equipment at the conclusion of the test.

4.3.3.5 Reliability and maintainability report. A report pertaining to reliability and maintainability shall be prepared after completion of the extended road test. The report shall contain the following information:

(a) All failures, maintenance, and other events shall be recorded and identified by accumulated operating time, distance, cycles, or position in the test procedure as appropriate. Test conditions during the failures or irregular operations identified shall be recorded.

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(b) Description of the engineering reasoning, and of any tests conducted to determine assignable causes for all failures or irregular operations identified.
(c) Description of the engineering reasoning behind any proposed corrections to be made on the production items, and the predicted effectiveness of those corrections.
(d) Test activity or contractor comments on item, features or requirements that, if modified, should improve the item.
(e) Estimates (within 5 percent of actual experienced) of manhours required for each maintenance and servicing action during the test. A brief description of the qualifications and experience of the personnel involved shall be included and shall be adequate to allow comparison to the personnel anticipated in similar field work.

4.3.4 Brake tests. Brake tests to determine conformance to requirements shall be conducted during the road test. The brake tests shall be conducted with the fully loaded semitrailer coupled to a truck tractor. The truck tractor shall have a gross vehicle weight rating (GVWR) and gross combination weight rating (GCWR) not exceeding that specified in table VI. The truck tractor shall be without excessive brake capacity. GVWR and GCWR shall be the maximum ratings on the tractor only and not necessarily the actual maximum GVW and GCW of the combination of test vehicles. Failure of the vehicle combination to stop within the required 12 200 mm (40 feet) from a speed of 32 km/h (20 mph) shall be cause for rejection.

TABLE VI. Maximum tractor characteristics.

Type of semitrailer	Max. tractor GVWR, kg (pounds)	Max. tractor GCWR, kg (pounds)
I	12 700 (28,000)	23 600 (52,000)
II	14 500 (32,000)	29 500 (65,000)
III	16 350 (36,000)	36 300 (80,000)
IV	24 950 (55,000)	45 360 (100,000)

4.3.5 Turning ability test. A truck-tractor shall be coupled to the semitrailer and driven through turns up to 90 degrees to the towed vehicle, right and left, to determine conformance to 3.3.1. At the conclusion of the turns in each direction, the tractor and the semitrailer shall be examined for evidence of binding, misalinement or other damage.

4.3.6 Parking brake test. To test parking brakes, the semitrailer, with rated payload, shall be placed on a grade, first headed up the grade and then headed down the grade, and tested to determine conformance to 3.10.1. These tests shall be conducted with the semitrailer disconnected from the towing vehicle and resting on the landing legs.

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4.3.7 Landing leg test. With the semitrailer disconnected from the towing vehicle and loaded as specified in 3.4, the landing legs shall be raised and lowered to vary the height of the upper fifth wheel over the landing gear's full range, to determine conformance to 3.12 and 3.12.1. The loaded semitrailer shall be coupled and uncoupled no less than 5 times and the landing legs shall be examined for binding and or misalinement.

4.3.8 Electrical test. An electrical test shall be conducted to determine that all vehicle lights are functioning. In addition, when an innerconnected 24-volt system is furnished, the circuits of the 12-contact receptacle shall be energized with not less than 28 volts dc regulated voltage for a time sufficient to stabilize the temperature of the resistor assembly for not less than 30 minutes in an ambient temperature of not less than 25°C (77°F). After the test, the resistor assembly, including resistors, wiring and adjacent components shall be examined for evidence of overheating, deterioration or damage.

4.3.9 Lifting and tiedown test. Lifting and tiedown provisions, markings, and the connecting structural members of the vehicle shall be inspected for conformance with MIL-STD-209. All lifting and tiedown provisions and the connecting structural members shall meet the criteria of MIL-STD-209. Test results that show the provisions do not meet the criteria of MIL-STD-209 shall constitute failure of this test.

4.3.10 Air transportability verification. The vehicle shall be inspected to determine that it conforms to the contractor's air transportability drawings and data as submitted to the Government for transportability approval. As a minimum, the following angles, dimensions and descriptions shall be checked against the Government approved contractor's drawings and data:

(a) Angle of approach
(b) Ramp breakover angle
(c) Angle of departure
(d) Height, longitudinal location and identification of highest component on semitrailer
(e) Dimensions and locations of any significant projections on semitrailer
(f) Curb weight distribution on each axle and kingpin
(g) Curb weight distribution on each axle and landing gear
(h) Front overhang
(i) Rear overhang
(j) Articulation of rear suspension, unloaded, each axle (curb weight)
(k) Rear axle spacing.

4.3.11 Certifications. The contractor shall certify that all wood requiring treatment in accordance with MIL-STD-1223 has been so treated. When an air suspension system is furnished, the contractor shall certify capability of the system to operate over the temperature range specified in 3.8.1.

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4.3.12 Production sample. Upon acceptance of the first production vehicle, it shall remain at the manufacturing facility as a production sample, and shall be the last vehicle shipped on the contract. The contractor shall maintain the vehicle in an as new condition for the duration of the contract. Release of the first production vehicle for shipment prior to completion of the contract shall be at the discretion of the local Government Quality Assurance Representative (QAR).

4.3.13 Failure. Failure of the first production vehicle to meet requirements of the contract shall be cause for the Government to refuse acceptance of all vehicles under the contract until corrective action has been taken.

4.4 Inspection of production vehicles. The contractor's inspection system shall, as a minimum, assure that the vehicle conforms to the physical and dimensional requirements and is capable of meeting the performance requirements contained herein.

5. PACKAGING

5.1 Vehicle processing. The vehicle shall be processed for shipment, from the manufacturer's plant to the initial receiving activity, in accordance with manufacturer's standard commercial practice.

6. NOTES

6.1 Intended use. The semitrailers covered by this specification are non-tactical vehicles (NTVs), intended for use by the Armed Forces for transporting general cargo. At times, cargo will be loaded on these semitrailers by forklifts having a 6350 kg (14,000 pound) axle load supported by 200 mm (8-inch) wide tires, with a tread width between wheels of 760 mm (30-inches).

6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number and date of this specification.
(b) Type, size and class of semitrailer required (see 1.2).
(c) Issue of DODISS to be cited in the solicitation and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
(d) Identification of appropriate military service for painting and marking (see 3.1.1.1 and 3.1.1.2).
(e) Independent 24-volt dc lighting system, if required (see 3.1.1.9.1).
(f) Interconnected 24-volt dc lighting system, if required (see 3.1.1.9.2).
(g) Frame stress at 33 percent of yield strength, in lieu of 50 percent, if required (see 3.6).
(h) For type IV semitrailer, the allowable axle load on the highways in the State(s) specified (see 3.7).

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(i) Sliding type axle, if required (see 3.7.1).
(j) Air suspension system, if required (see 3.8.1).
(k) Air/hydraulic powered landing gear, if required (see 3.12.2).
(l) Container securement devices, if required (see 3.16.5).
(m) Roof bows and tarpaulin for class 1 slat side semitrailer, if required (see 3.20).
(n) If roof bows and tarpaulin are not required for class 2 panel side semitrailer (see 3.20).

6.3 Subject term (key word) listing.

Air transportability
Container securement devices
Forklifts
Hardwood
Non-tactical vehicle (NTV)
Panel sides
Road test
Stake racks
Tarpaulin
Welder's qualifications.

Custodians:
Army - AT
Navy - YD
Air Force - 99

Review Activity:
Air Force - 84

User Activity:
Navy - MC

Preparing Activity:
Army - AT

Project No. 2330-0128

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER KKK-S-002769A	2. DOCUMENT DATE (YYMMDD)

3. DOCUMENT TITLE
SEMITRAILERS, STAKE: MODIFIED COMMERCIAL

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME *(Last, First, Middle Initial)*	b. ORGANIZATION	
c. ADDRESS *(Include Zip Code)*	d. TELEPHONE *(Include Area Code)* (1) Commercial (2) AUTOVON *(If applicable)*	7. DATE SUBMITTED *(YYMMDD)*

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE *(Include Area Code)*	
JAMES L. BRUBKAER	(1) Commercial (313)574-6407	(2) AUTOVON 786-6407
c. ADDRESS *(Include Zip Code)* US Army Tank Automotive Command Warren, Michigan 48397-5000 AMSTA-UED	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON: 289-2340	

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